

Sion King

904-609-1735 | sionjking@gmail.com | [linkedin.com/in/sionjking](https://www.linkedin.com/in/sionjking) | github.com/sionjk

EDUCATION

University of Central Florida

Bachelor of Science in Statistics

Orlando, FL

Expected December 2027

EXPERIENCE

College of Sciences Student Ambassador

University of Central Florida

August 2026 – Present

Orlando, FL

- **Facilitate interactive meetings** and social events for College of Sciences students to **foster community building, peer connection**, and a positive campus experience
- Serve as a liaison between College of Sciences students and university leadership by **communicating student perspectives** and **supporting college-wide initiatives**
- **Engage directly with university administration**, faculty, and alumni, to support college relations

Finance Team @ KnightHacks UCF

University of Central Florida

July 2026 – Present

Orlando, FL

- Supported **budgeting** and **financial planning** for Knight Hacks, helping manage a **\$100K+ annual budget** across sponsorship revenue, event programming, and operating expenses
- Refined the **procurement pipeline** by standardizing purchase requests, approval tracking, and reimbursement documentation, **reducing** average processing time by **34 days**
- Tracked departmental spending across **6 teams** and provided recurring budget updates, helping leadership identify variances and **prevent cost overruns**

PROJECTS

Halo

- Won **1st of 77** projects for **Waymo** and **1st of 34** for **State Farm** at **ShellHacks 2026** with a blind-spot detection bike helmet, leading its computer vision, sensor fusion, and real-time visualization
- Assessed **real-time sensor data** using **Python** to estimate vehicle distance, trajectory, passing clearance, and time-to-collision
- Applied **median filtering** and **temporal thresholds** to reduce measurement noise and false hazard alerts

Happiness and Social Media Multiple Linear Regression Model

- Examined **500 observations** across **13 predictors** using **R** to examine relationships between social media use, behavioral factors, and reported happiness
- Validated a multiple linear regression model using **10-fold cross-validation** to assess predictive performance, achieving an $R^2 = 0.639$, $RMSE = 0.920$, and $MAE = 0.750$
- Evaluated model assumptions including **linearity**, **normality**, and **homoscedasticity** to assess **model validity**
- Applied **response transformation** and **model diagnostics** to improve model fit to $R^2 = 0.665$, identifying 3 statistically significant predictors: sleep quality, stress level, and age

Graduate School Multiple Linear Regression Model

- Analyzed **400 graduate school admission records** across **6 predictors** using **R** to identify academic factors associated with admission probability
- Developed a multiple linear regression model explaining **79.8% of variation in admission outcomes**, achieving an adjusted R^2 of 0.796 across 400 observations
- Constructed 95% confidence and prediction intervals and compared full and **reduced regression models** to evaluate **prediction uncertainty** and **model efficiency**
- Reduced model from **6 to 4 significant predictors** while retaining **99.9% of its explanatory power**, identifying GRE, TOEFL, LOR, and CGPA as strongest final predictors

TECHNICAL SKILLS

Relevant Coursework: Stat I-III, Big Data Analytics, Foundations of Data Science and AI, Data Science I

Statistical Methods: Multiple Linear Regression, Statistical Inference, Hypothesis Testing, Confidence & Prediction Intervals, Correlation Analysis, Regression Diagnostics, Model Selection

Data Analysis: Data Cleaning, Data Visualization, Predictive Modeling, Model Evaluation

Languages: R, Python, SQL, C

Tools & Libraries: Excel, Pandas, NumPy, Matplotlib, RStudio, Jupyter Notebook, Git/GitHub